

General Knowledge

Indian Geography

STUDY NOTES

**Energy Resources in India - Nuclear
Power plants**



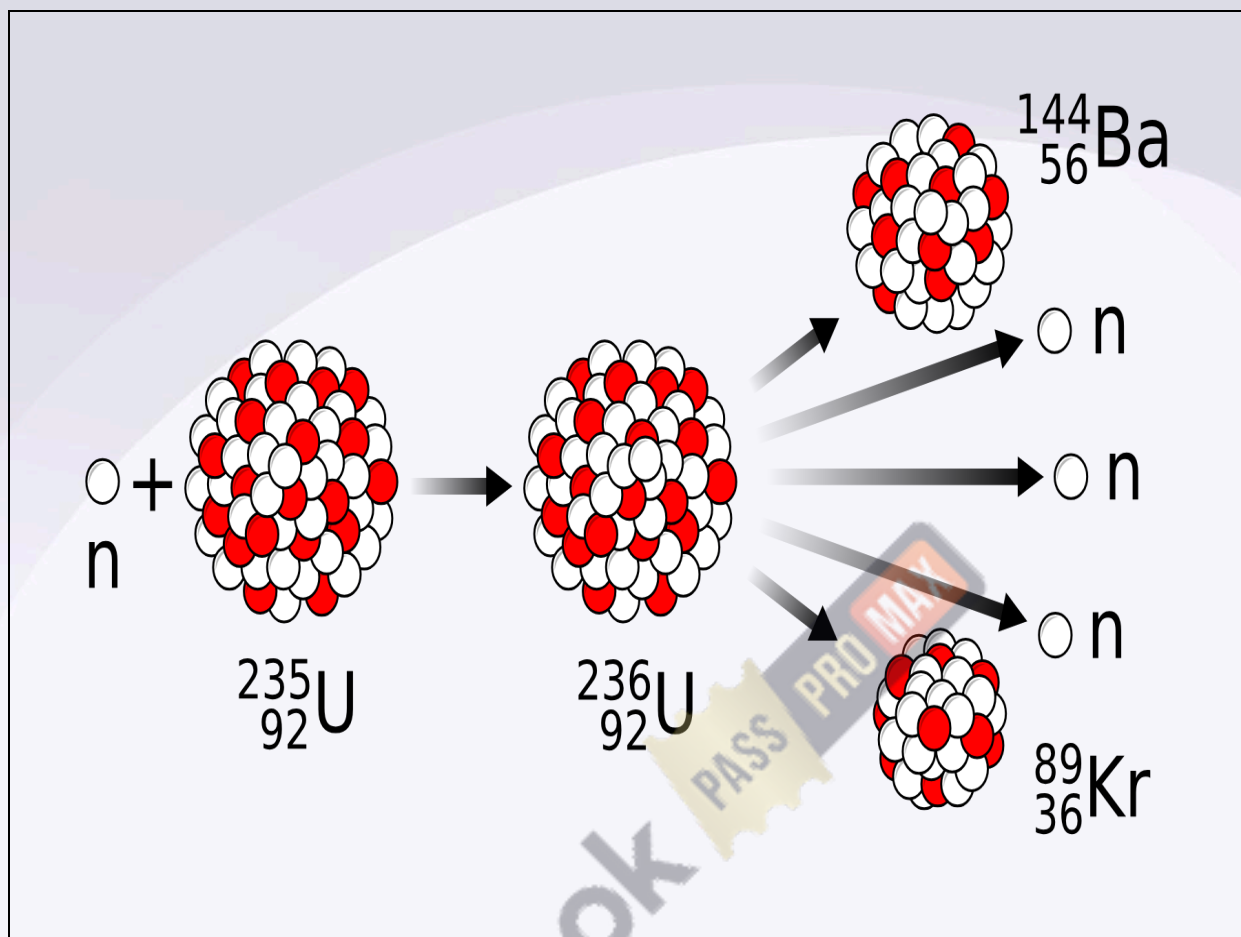
NUCLEAR ENERGY IN INDIA

Nuclear energy plays a pivotal role in addressing the country's growing energy demands. India is one of the few countries that have successfully harnessed nuclear energy, producing electricity using indigenously developed nuclear reactors. This power source not only helps in generating electricity but also contributes significantly to the country's long-term energy security.



Definition of Nuclear Energy

Nuclear energy is produced through a **nuclear fission process** where atoms are split to generate heat, which turns water into steam, driving a turbine and generating electricity. Nuclear reactors are the core machinery for this process.



India's Nuclear Program and Energy Development

- **Initiation of India's Nuclear Program (1950s):**

- India's nuclear program began in the 1950s with a primary focus on ensuring **energy security**.
- The **Three-Stage Nuclear Power Programme**, formulated by **Dr. Homi Bhabha**, laid the foundation for the country's nuclear energy journey.

- **The Atomic Energy Act (1962):**

- To advance the use of **Uranium** and **Thorium** as nuclear fuels, India passed the **Atomic Energy Act** in 1962, enabling the development of nuclear technology.

- **Nuclear Tests Conducted by India**

- **First Peaceful Nuclear Explosion (PNE):**

- i. **Year:** 1974
 - ii. **Location:** Pokhran Test Range, Rajasthan
 - iii. **Codename:** Smiling Buddha
 - iv. **Significance:** Marked India's entry into the global nuclear stage as a nuclear power.

- **Pokhran-II (Operation Shakti):**

- v. **Year:** May 1998
 - vi. **Location:** Pokhran Test Range, Rajasthan
 - vii. **Codename:** Operation Shakti
 - viii. **Tests Conducted:** Five nuclear devices were detonated during this operation.
 - ix. **Significance:** Demonstrated India's ability to develop both fission and thermonuclear (fusion) devices, highlighting India's diverse nuclear capabilities.

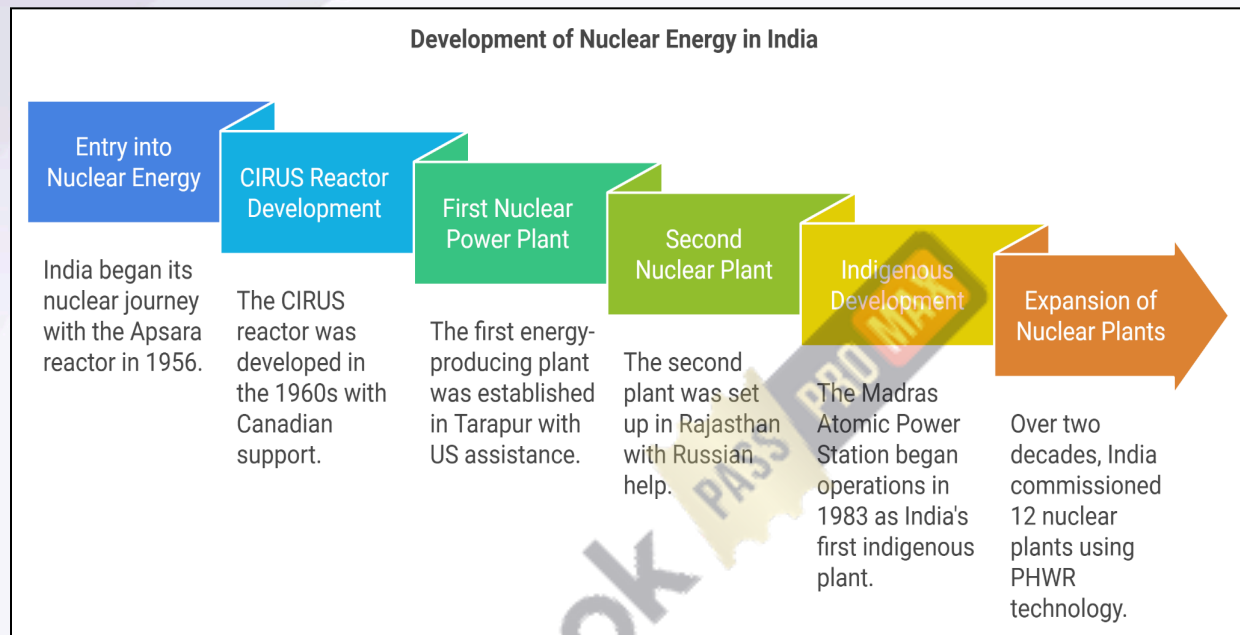
- **International Backlash and Nuclear Non-Proliferation Treaty (NPT):**

- Despite widespread international support for the peaceful use of nuclear energy, **India faced restrictions** on developing nuclear strategic and military capabilities.
 - India refused to sign the **Nuclear Non-Proliferation Treaty (NPT)** due to its discriminatory nature, which led to limited access to global nuclear resources.

- **Nuclear Suppliers Group (NSG) Waiver (2008):**

- In **2008**, after years of challenges, the **Nuclear Suppliers Group (NSG)** granted India a **waiver**, allowing the country to engage in **nuclear commerce for civilian purposes**.

About Nuclear Energy in India



- **India's Entry into Nuclear Energy (1956):**

- India entered the **nuclear age** in **August 1956** with the installation of the **Apsara nuclear reactor**, operating initially with **USA-supplied fuel**.

- **CIRUS Reactor Development (1960s):**

- In the **1960s**, India built the **CIRUS reactor** with **Canada's support** for **research purposes**, which contributed to innovations in **medical diagnostics** and **industrial applications**.

- **India's First Nuclear Power Plants:**

- The first **energy-producing nuclear power plant** was established in **Tarapur**, with **USA's assistance**.

- The **second nuclear plant** was set up in **Rajasthan** with the help of **Russia**.

- **Indigenous Nuclear Development (1983):**

- India's **first indigenously developed nuclear power plant** was the **Madras Atomic Power Station (MAPS)**, which began operations in **1983**.

India's Position in Global Nuclear Energy Production

India ranks **third globally** in terms of **electricity production**, with nuclear energy being the **fifth-largest source of electricity** in the country. India's nuclear energy contribution remains essential for its energy transition plans, contributing toward its net-zero economy goals.

Potential of Nuclear Energy in India

India is progressing towards increasing its nuclear energy production, aiming to boost its atomic power contribution from **3.2% to 5% by 2031**. This will help India meet its growing energy needs while transitioning to a cleaner, more sustainable energy future.

List of Operational Nuclear Power Plants in India

Name of Nuclear Power Plant	Location	Capacity (MW)
Kakrapar Atomic Power Station – 1993	Gujarat	440
Madras Atomic Power Station (Kalpakkam) – 1984	Tamil Nadu	440
Narora Atomic Power Station – 1991	Uttar Pradesh	440
Kaiga Nuclear Power Plant – 2000	Karnataka	880
Rajasthan Atomic	Rajasthan	1,180

Power Station – 1973		
Tarapur Atomic Power Station – 1969	Maharashtra	1,400
Kudankulam Nuclear Power Plant – 2013	Tamil Nadu	2,000

India's Nuclear Power Capacity

India currently has over **22 nuclear reactors** spread across **7 power plants** that generate a total of **6780 MW of electricity**.

Plans for Expansion

To meet energy demands, India plans to further expand its nuclear power capacity by constructing **12 new nuclear power reactors by 2024**, with the intention of enhancing its energy portfolio while reducing dependency on fossil fuels.

Nuclear Energy Resources in India

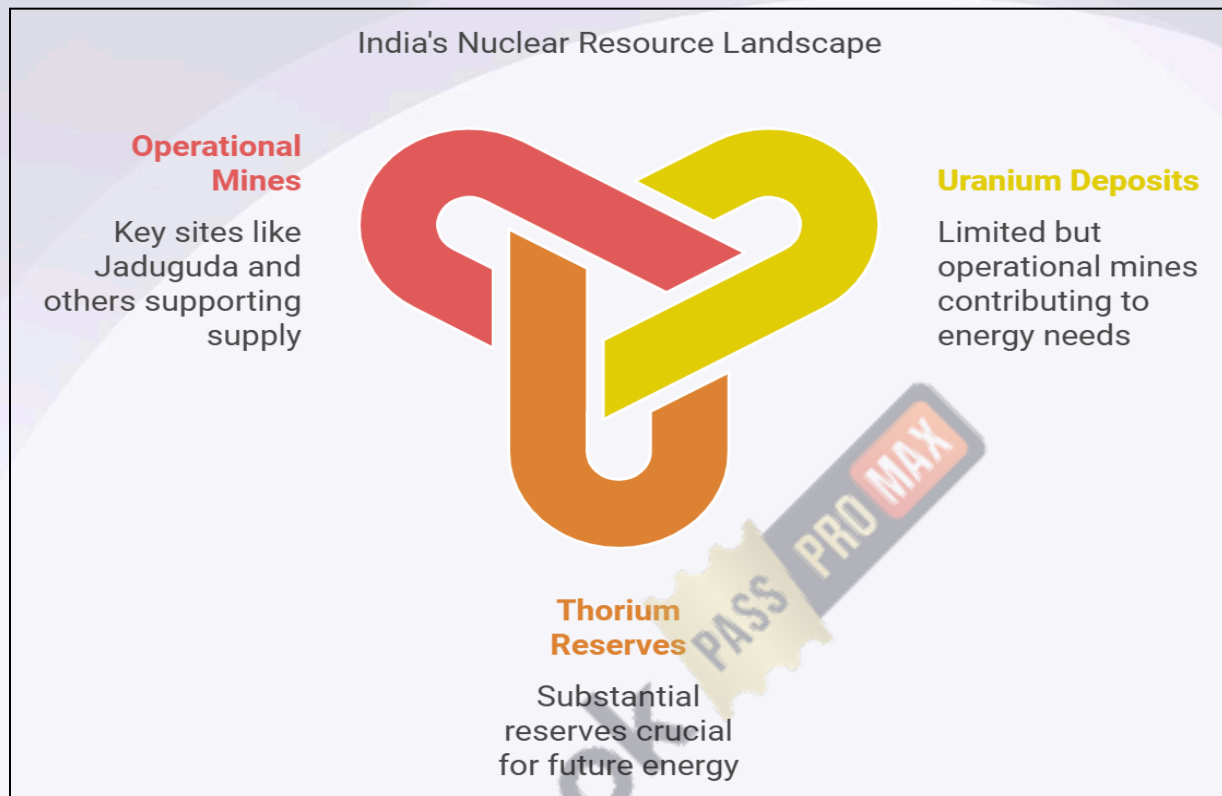
India faces challenges in fulfilling its growing energy demands due to limited domestic uranium resources, but it holds substantial thorium reserves, which could be crucial for the country's nuclear future.

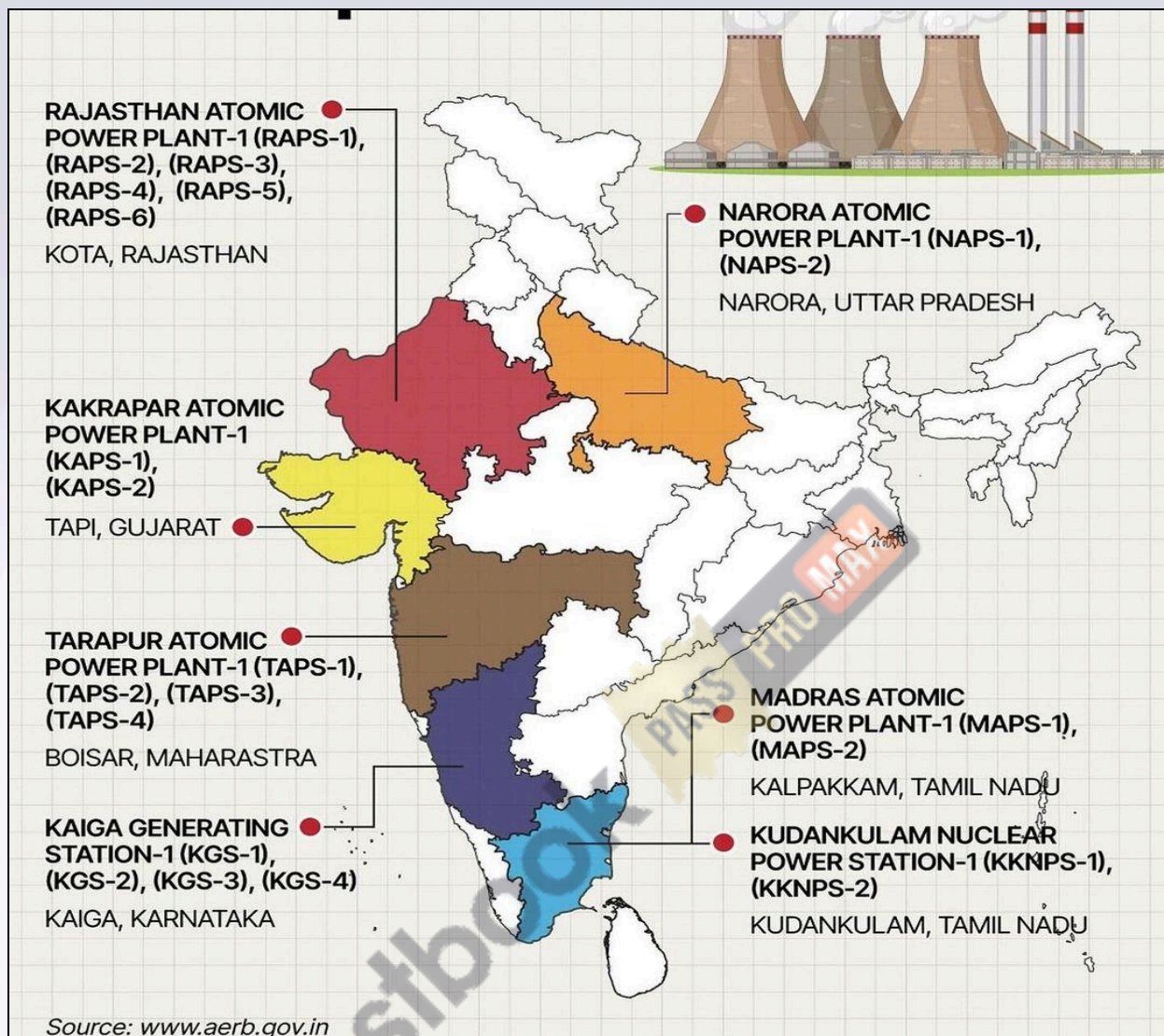
Uranium Deposits

Uranium is the primary fuel used in most nuclear reactors, and although India possesses limited uranium deposits, it has several operational mines:

- **Jaduguda:** The first uranium deposit discovered in India, located in the Singhbhum Thrust Belt (Jharkhand).
- **Other Mines:** Mines like Bhatin, Narwapahar, and Turamdih continue to operate, contributing significantly to the uranium supply.

Additional uranium occurrences are found in Andhra Pradesh, Meghalaya, Rajasthan, and Chhattisgarh.

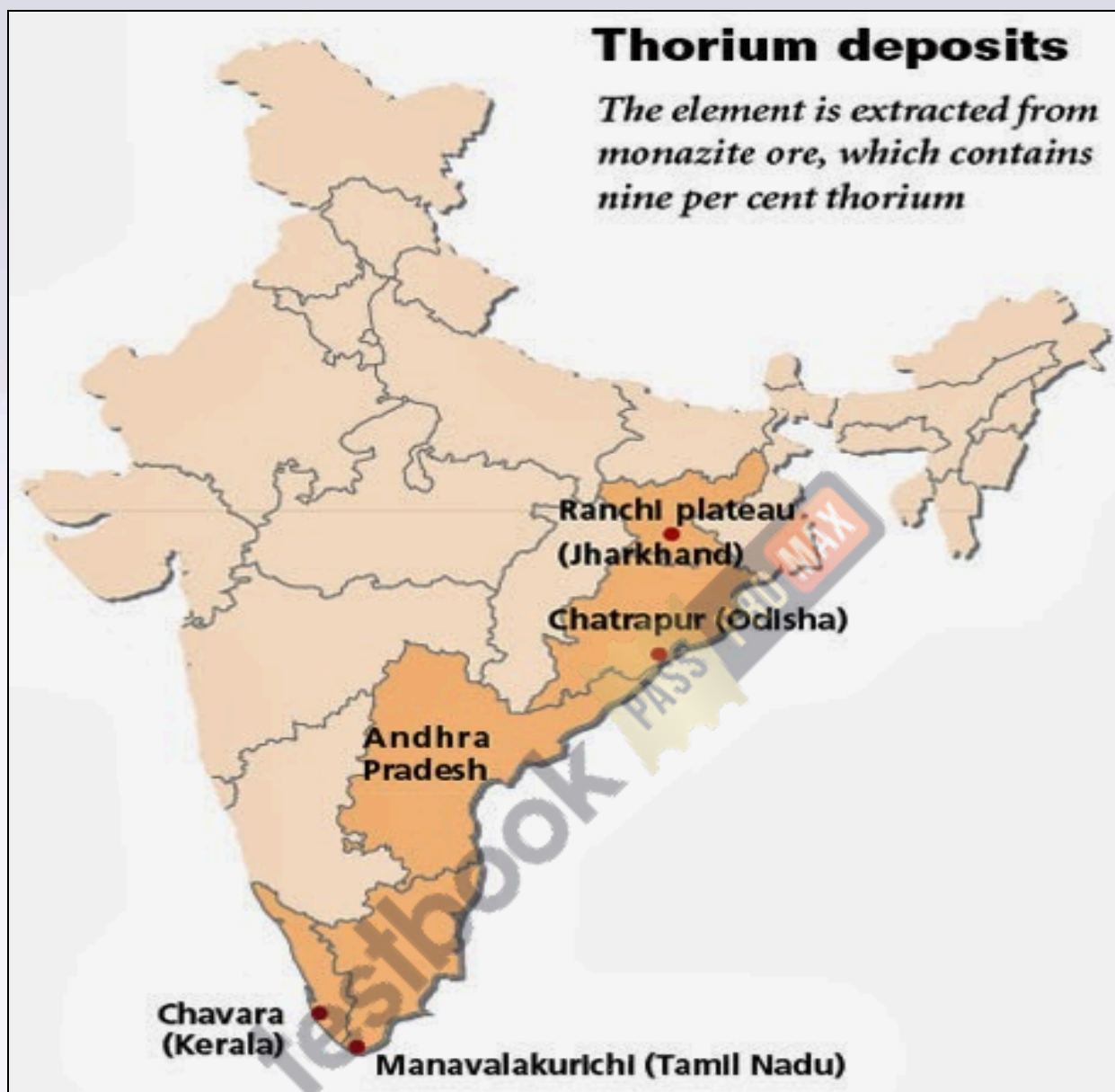




Thorium Deposits

India is one of the leading nations with abundant thorium resources, particularly in the form of monazite sands located along the east and west coasts, with a concentration of thorium in Kerala sands. Thorium is a promising alternative fuel for nuclear reactors.

- **Main Producing States:** Rajasthan, Tamil Nadu, Jharkhand, Bihar, and Kerala.
- **Kerala's Monazite Sands:** Contains significant thorium reserves.



Nuclear Power Plants in India

India operates several types of reactors, and the country is continuing to advance its nuclear power infrastructure. Nuclear power plants in India are largely composed of Pressurized Heavy Water Reactors (PHWRs) and Light Water Reactors (LWRs).

Existing Nuclear Reactors

India operates 4 Light Water Reactors (LWRs) and 19 Pressurized Heavy Water Reactors (PHWRs).

- **Kakrapar Nuclear Power Plant (Gujarat):** India's first large-scale indigenous 700 MWe reactor, operating at full capacity.
- **Reactors under IAEA Safeguards:** India's fourteen nuclear reactors are under IAEA safeguards for international oversight.

Nuclear Power Plant Development

India aims to construct 12 new nuclear reactors by 2024 as part of its strategy to meet future energy requirements.

India's Three-Stage Nuclear Power Program

Dr. Homi Bhabha, the father of the Indian nuclear program, in 1950s, formulated a three-stage nuclear power program to ensure the country's long-term energy independence.

Stage 1: Pressurized Heavy Water Reactors (PHWRs)

- **Fuel:** Natural uranium.
- **Byproduct:** Plutonium-239.
- **Goal:** To utilize uranium efficiently while avoiding uranium enrichment, which reduces waste and provides for refueling without reactor shutdown.

Stage 2: Fast Breeder Reactors (FBRs)

- **Fuel:** Plutonium-239 is produced from the first stage.
- **Byproduct:** More uranium-239 for subsequent use.
- **Advantages:** FBRs are energy-efficient, as they recycle nuclear waste from the first stage.

- **Cooling Agent:** Liquid sodium (which can react explosively with water and air).
- **Safety Concerns:** Although FBRs are effective, their containment structures are not as robust as conventional reactors.

Stage 3: Thorium-Based Reactors (TBRs)

- **Fuel:** Thorium, which will generate uranium-233 upon irradiation.
- **Goal:** To move towards utilizing abundant thorium, making nuclear energy more sustainable and greener.
- **Advantages:** Thorium is safer and greener than uranium; it also produces fewer harmful byproducts.

Key Institutions Driving Nuclear Energy in India

India's nuclear energy sector is governed by several prominent organizations focused on research, safety, regulation, and the operation of nuclear plants, which play a crucial role in advancing the country's energy goals.

Bhabha Atomic Research Centre (BARC)

- Operates under the Department of Atomic Energy (DAE).
- **Primary Objective:** Ensures the peaceful use of nuclear energy within India.
- **Main Role:** Supervises all aspects of nuclear power production in India.

Atomic Energy Regulatory Board (AERB)

- Established in 1983 under the Atomic Energy Commission.
- **Core Function:** Regulates nuclear safety and ensures that proper safety guidelines are followed at nuclear facilities across India.

Nuclear Power Corporation of India (NPCIL)

- The exclusive operator and owner of all nuclear power plants in India, except for the Prototype Fast Breeder Reactor (PFBR).
- **Responsibilities:**
 - Responsible for designing, constructing, commissioning, and managing the operation of nuclear reactors.

Atomic Energy Commission (AEC)

- Governs the Department of Atomic Energy (DAE), which is directly overseen by the Prime Minister of India.
- **Key Functions of AEC:**
 - Promotes research in atomic science.
 - Trains nuclear scientists within the country.
 - Supports nuclear research in laboratories under DAE.

